

Workshop 4

1. Are either, both, or neither of the following two models GLMs? Explain.

$$\log(y_i) = \beta_0 + \beta_1 x_i, \quad y_i \sim \text{gamma},$$

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2. Suppose that y_i represents the number of bacterial cells in a sample from a petri dish and x_i is the amount of nutrient in the dish. If it is expected that:

$$\mathbb{E}(y_i) = a^2 + 2bax_i + b^2x_i^2,$$

write down a suitable GLM for the data (y_i, x_i) .

3. The following table classifies a random sample of women and men according to their belief in the afterlife:

	Believer	Non-Believer
Female	435	147
Male	375	134

The data come from the US General Social Survey (1991), and the ‘non-believer’ category includes ‘undecideds’. Are there differences between males and females in the holding of this belief? A null model for the data would be that belief does not depend on gender, in which case, if y_i represents the observed count in one of the table’s cells, then we could model the expected number of counts as

$$\mu_i \equiv \mathbb{E}(y_i) = n\gamma_k\alpha_j \text{ if } y_i \text{ is data for gender } k, \text{ and faith } j.$$

Here n is the total number of people sampled, γ_1 and γ_2 are the proportion female and male in the population sampled, and α_1 and α_2 are the proportion of believers and non-believers in the population sampled, respectively.

For practical modelling purposes, it turns out that an appropriate response distribution to use here is the Poisson distribution. For the moment, show that this model can be turned into a GLM using the log link. Write out the full model matrix for the model, and then a version where the parameters are identifiable.

4. In January 1986, the space shuttle *Challenger* exploded shortly after takeoff. A subsequent investigation eventually focused on the possibility that O-rings in the fuel tanks had failed due to the unusually low launch temperature (31°F), allowing a fuel leak. It turned out that data on O-ring failures in previous launches was available for various launch temperatures. Below is a plot showing the number of failed O-rings (out of a total of six) in launches at different temperatures. The first few rows of the dataset are also shown.



5. Show that the exponential distribution with probability density function given by

is in the exponential family of distributions. Further show, using the properties of the exponential family of distributions, that $\mathbb{E}(Y) = 1/\lambda$ and $\text{var}(Y) = 1/\lambda^2$.

- $$f(y; p) = p^y(1 - p)^{1-y}, \quad y \in \{0, 1\}, \quad p \in [0, 1].$$

- $$f(y; \pi, n) = \binom{n}{ny} \pi^{ny} (1 - \pi)^{n - ny}, \quad y = 0, \frac{1}{n}, \frac{2}{n}, \dots, 1.$$

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8. (a) Show that the Inverse Gaussian distribution with density given by

$$f(y; \mu, \gamma) = \sqrt{\frac{\gamma}{2\pi y^3}} \exp \left\{ -\frac{\gamma(y - \mu)^2}{2\mu^2 y} \right\}, \quad y > 0, \mu > 0, \gamma > 0,$$

where μ is the mean and γ is a scaling parameter, can be written in the form

$$f(y; \theta, \phi) = \exp \left\{ \frac{y\theta - b(\theta)}{a(\phi)} + c(y, \phi) \right\},$$

and determine θ , $b(\theta)$, $a(\phi)$, and $c(y, \phi)$.

- (b) Show that $\mathbb{E}(Y) = \mu$ and $\text{var}(Y) = \frac{\mu^3}{\gamma}$.

- (c) What is the canonical link function for this distribution?

9. Show that the t distribution, which is suitable for modelling heavy tail data and whose probability density function is given by

$$f(y; \nu) = \frac{\Gamma(\frac{\nu+1}{2})}{\sqrt{\nu\pi}\Gamma(\frac{\nu}{2})} \left(1 + \frac{y^2}{\nu}\right)^{-\frac{\nu+1}{2}}, \quad y \in \mathbb{R}, \quad \nu > 0,$$

is not in the exponential family.

10. In a study of the presence of tumors in animals, suppose

$$Y_i \stackrel{\text{ind}}{\sim} \text{Poisson}(\mu_i), \quad \log(\mu_i) = \mathbf{x}_i^\top \boldsymbol{\beta}, \quad i = 1, \dots, n.$$

However, the observed response, Z_i , merely indicates whether each Y_i is positive, $Z_i = I(Y_i > 0)$. Define $\pi_i = \Pr(Z_i = 1)$. Then, the Z_i s follow a binomial GLM

$$Z_i \stackrel{\text{ind}}{\sim} \text{Bin}(1, \pi_i), \quad i = 1, \dots, n.$$

Show that the link function for this binomial model is given by

$$\log(-\log(1 - \pi_i)) = \mathbf{x}_i^\top \boldsymbol{\beta}.$$

11. Let y_i be independent random variables each with mean μ_i and variance $V(\mu_i)\phi$, where $\mu_i = g^{-1}(\mathbf{x}_i^\top \boldsymbol{\beta})$, g is a smooth monotonic function, $\boldsymbol{\beta}$ a vector of parameters, and $\mathbf{X}$ a design matrix of known entries. Let $z_i = g'(\mu_i)(y_i - \mu_i) + \mathbf{x}_i^\top \boldsymbol{\beta}$. Find the expected value and covariance matrix of the vector $\mathbf{z} = (z_1, \dots, z_n)^\top$.

12. This question is about getting a feel for the IRLS method used to fit GLMs.

- (a) Find an expression for the value of $\boldsymbol{\beta}$ minimizing

$$\sum_{i=1}^n w_i (z_i - \boldsymbol{\beta} x_i)^2$$

in terms of w_i , z_i , and x_i .

(b) Consider fitting the following GLM

$$\mu_i \equiv \mathbb{E}(y_i) = \beta x_i, \quad y_i \sim \text{Poi}(\mu_i),$$

to the data:

x	1	2	3	4	5
y	0	1	3	7	4

Apply two steps of the IRLS scheme, by hand, in order to obtain an MLE of β . You will need to modify the default starting values for the IRLS, by changing the initial value used for $\hat{\mu}_1$ (why?): I suggest starting with $\hat{\mu}_1 = 1$.

(c) Obtain an estimate of the standard error of $\hat{\beta}$.